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Project Number: #0

Project Name: Simple OpenMP Experiment

Project Report # 0

- a. Tell me what machine you ran this on

ARM M2 Mac OS

- b. What performance results did you get?

1. For 1 thread, Peak performance = 180.37 MegaMults/Sec

```
> ./program
OpenMP version 201811 is supported here
For 1 threads, Peak Performance = 180.37 MegaMults/Sec
```

2. For 4 threads, Peak Performance = 348.83 MegaMults/Sec

```
> ./program
OpenMP version 201811 is supported here
For 4 threads, Peak Performance = 348.83 MegaMults/Sec
```

- c. What was your 1-thread-to-4-thread speedup?

My Speedup = $348.83/180.37 = 1.933969$ megaMult/Sec

- d. Your 1-thread-to-4-thread speedup should be less than 4.0. Why do you think it is this way?

I think my 1-thread-to-4-thread speedup should be less than 4.0 because threads share the same data area, text area, and heap. With multiple threads accessing shared resources such as memory, cache, or I/O devices, contention for these resources can occur. This contention can lead to delays as threads wait for access to critical sections of code or shared data structures, thereby reducing the overall performance improvement. Moreover, according to Amdahl's Law, the speedup of a parallel program is limited by the fraction of the program that cannot be parallelized. Even if a program contains parallelizable sections, there may still be sequential portions that limit the overall speedup achievable by parallelism. That's why the speedup from 1 thread to 4 threads is typically less than linear.

- e. What was your Parallel Fraction, F_p ? (Hint: it should be less than 1.0, but not much less.)

$$F_p = 0.643905$$